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Experiment No 9

14/10

Q1 Explain or define index and write its applications

⇒ An index is a database object that helps the system find rows in a table quickly without scanning every row. It works like the index at the back of a book. It stores the values of one or more columns in a sorted structure, usually a B tree, along with pointers to the actual rows. Indexes make reading faster, but they take extra storage and slow down inserts, updates and deletes slightly, since the index must also be maintained.

Applications :-

1) Speeding up SELECT queries that search on a particular column using the WHERE clause
2) Making the JOIN operations faster by indexing the columns used to join tables
3) Speeding up ORDER BY and GROUP BY operations on large tables.
4) Enforcing uniqueness on a column through a unique index, such as email or roll number.
5) Improving performance in large databases such as banking, library and e-commerce systems where millions of rows are searched often.

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Q2 Write a syntax for showing and creating index.

⇒ 1) Creating an Index :-

CREATE INDEX idxname ON tablename(columnname);

2) Creating a unique Index :-

CREATE UNIQUE INDEX idxname ON tablename(columnname);

3) Creating a composite index on more than one column :-

CREATE INDEX idxname ON tablename(column1, column2)

4) Showing indexes of a table :-

SHOW INDEX FROM tablename;

5) Dropping an Index :-

DROP INDEX idxname ON tablename;

Example :-

CREATE INDEX idxstudentname ON student(name);
SHOW INDEX FROM student;

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